

IMO Shortlist 2012 — C1

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Several positive integers are written in a row. Iteratively, Alice chooses two adjacent numbers x and y such that $x > y$ and x is to the left of y , and replaces the pair (x, y) by either $(y + 1, x)$ or $(x - 1, x)$. Prove that she can perform only finitely many such iterations.

Solution

Let the positive integers be $a_1, a_2, \dots, a_n$. Notice that if (a_n) becomes increasing, then Alice will no longer be able to perform any operations.

Let $x > y$ be adjacent numbers with x on the left of y . Since $\max(y + 1, x - 1) \leq x$, once we do the operation on (x, y) the number on the right will be $\geq$ the number on the left. Suppose FTSOC that it's possible to perform an infinite number of iterations.

Claim 1: Let M be the largest number such that $a_M = \max(a_n)$. Eventually it must end up at the end of the list.

Proof: Notice that the largest number in the list can only ever move right under this operation. Suppose FTSOC that the claim is false. That is, after a number of operations the number a_M always stays in some fixed position $k < n$ in the list. Consider the lists $a_1, a_2, \dots, a_{k-1}$ and $a_{k+1}, a_{k+2}, \dots, a_n$. Now since claim 1 is false the largest number in each list will have some fixed position that isn't the end of the list after some number of operations also. If we keep repeating this argument then eventually we will reach a point where all the numbers in (a_n) have fixed positions and so no more operations can be performed, a contradiction. $\square$

So due to claim 1, the largest number a_M will end up at the end of the list after finitely many operations. Now since it's the largest number it can no longer move. Now consider the list $a_1, a_2, \dots, a_{n-1}$. Due to claim 1, the largest number in this new list will eventually end up at the end of the list after finitely many operations also. If we repeat this argument then we will show that (a_n) will eventually become sorted in increasing order after finitely many operations, and so after this point Alice will no longer be able to perform any more operations, a contradiction. $\blacksquare$